EXPERIMENT NO 9

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INPUT CODE

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#include <limits.h>
#include <stdbool.h>
#include <stdio.h>

#define V 5

int minKey(int key[], bool mstSet[]) {
    int min = INT_MAX;
    int min_index = -1;

    for (int v = 0; v < V; v++) {
        if (mstSet[v] == false && key[v] < min) {
            min = key[v];
            min_index = v;
        }
    }
    return min_index;
}

void printMST(int parent[], int graph[V][V]) {
    printf("Edge \tWeight\n");
    for (int i = 1; i < V; i++) {
        printf("%d - %d \t%d \n", parent[i], i, graph[parent[i]][i]);
    }
}

void primMST(int graph[V][V]) {
    int parent[V];
    int key[V];
    bool mstSet[V];

    for (int i = 0; i < V; i++) {
        key[i] = INT_MAX;
        mstSet[i] = false;
```

```
    }

    key[0] = 0;
    parent[0] = -1;
    for (int count = 0; count < V - 1; count++) {
        int u = minKey(key, mstSet);
        if (u == -1) break;
        mstSet[u] = true;
        for (int v = 0; v < V; v++) {
            if (graph[u][v] && mstSet[v] == false && graph[u][v] < key[v]) {
                parent[v] = u;
                key[v] = graph[u][v];
            }
        }
    }
    printMST(parent, graph);
}

int main() {
    int graph[V][V] = {
        { 0, 2, 0, 6, 0 },
        { 2, 0, 3, 8, 5 },
        { 0, 3, 0, 0, 7 },
        { 6, 8, 0, 0, 9 },
        { 0, 5, 7, 9, 0 }
    };

    primMST(graph);
    return 0;
}
```

OUTPUT

